Algebraic Topology

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0 Abstract Algebra Prerequisites

0.1 Group Theory Basics

Definition 0.1.1. Group

A **group** $\langle G, * \rangle$ is a set G, closed under a binary operaton $*$, which satisfies

- associativity of $*$
- existence of identity element e
- existence of inverse element a' of a for all $a \in G$

Definition 0.1.2. Homomorphism and Isomorphism

Let $\langle G_1, *_1 \rangle$ and $\langle G_2, *_2 \rangle$ be groups and $f : G_1 \to G_2$. We say that f is a **group isomorphism** if

- f is bijective.
- For all $a, b \in G$, $f(a *_1 b) = f(a) *_2 f(b)$.

f is a **group homomorphism** if the second condition is satisfied.

Definition 0.1.3. Permutation and Symmetric Group

Let $A \neq \varnothing$. A **permutation** of A is a bijection $\sigma \colon A \to A$. The **symmetric group** on A is

$$S_A = \mathrm{Sym}(A) = \{\sigma \colon A \to A \mid \sigma \text{ is bijective}\}$$

For $A = \{1, 2, \ldots, n\}$, denote $S_n = S_{\{1,\ldots,n\}}$ with order $|S_n| = n!$.

Definition 0.1.4. Cycle and Transposition

A k**-cycle** $(a_1\ a_2\ \cdots\ a_k) \in S_n$ $(k \geq 2)$ is defined by

$$\sigma(a_i) = a_{i+1}\ (1 \leq i < k), \quad \sigma(a_k) = a_1, \quad \sigma(x) = x\ \forall x \notin \{a_1, \ldots, a_k\}\,.$$

A 2-cycle is a **transposition**. Inverses satisfy:

$$(a_1\ a_2\ \cdots\ a_k)^{-1} = (a_k\ a_{k-1}\ \cdots\ a_1).$$

Proposition 0.1.5. Let $\mathrm{supp}(\sigma) = \{x \in \{1, \ldots, n\} \mid \sigma(x) \neq x\}$.

(a) **Disjoint Commutativity**: $\mathrm{supp}(\sigma) \cap \mathrm{supp}(\tau) = \varnothing \implies \sigma\tau = \tau\sigma$.

(b) **Disjoint Cycle Decomposition**: $\forall \pi \in S_n$, $\exists!$ partition of moved elements into disjoint cycles $c_1, \ldots, c_m$ such that

$$\pi = c_1 c_2 \cdots c_m.$$

Definition 0.1.6. General Linear Group

For $n \in \mathbb{Z}^+$,

$$\mathrm{GL}(n, \mathbb{R}) = \{A \in M_n(\mathbb{R}) \mid \det(A) \neq 0\} = M_n(\mathbb{R})^\times.$$

$(\mathrm{GL}(n, \mathbb{R}), \cdot)$ is nonabelian for all $n \geq 2$.

Definition 0.1.7. Dihedral Group

For $n \geq 3$, the **dihedral group** D_n is the isometry group of a regular n-gon in $\mathbb{R}^2$:

$$D_n = \langle \rho, \mu \mid \rho^n = e,\ \mu^2 = e,\ \mu\rho\mu = \rho^{-1} \rangle.$$

Theorem 0.1.8. Let $n \geq 3$. Then $|D_n| = 2n$, with canonical form:

$$D_n = \left\{ \rho^k, \mu\rho^k \mid 0 \leq k < n \right\},$$

governed by:

$$\rho^k \mu = \mu\rho^{-k} = \mu\rho^{n-k}, \quad (\rho^k)^{-1} = \rho^{n-k}, \quad (\mu\rho^k)^{-1} = \mu\rho^k.$$

Definition 0.1.9. Subgroup

Let $(G, \cdot)$ be a group. $H \subseteq G$ is a **subgroup** $(H \leq G)$ if

$$H \neq \varnothing \quad \text{and} \quad (H, \cdot|_{H \times H}) \text{ is a group.}$$

$H < G$ denotes a proper subgroup $(H \neq G)$. $H = \{e\}$ is the **trivial subgroup**; $H = G$ is the **improper subgroup**.

Theorem 0.1.10 (Subgroup Criteria)**.** Let G be a group and $H \subseteq G$.

(a) **Three-Step Test**:

$$H \leq G \iff (\forall a, b \in H \implies ab \in H) \wedge (e \in H) \wedge (\forall a \in H \implies a^{-1} \in H).$$

(b) **One-Step Test**:

$$H \leq G \iff H \neq \varnothing \wedge (\forall a, b \in H \implies ab^{-1} \in H).$$

(c) **Finite Subgroup Test**: If $|H| < \infty$ and $H \neq \varnothing$,

$$H \leq G \iff (\forall a, b \in H \implies ab \in H).$$

Example 0.1.11 (Standard Subgroups)**.**

(a) **Center**: $Z(G) = \{z \in G \mid zg = gz\ \forall g \in G\} \leq G$ is abelian.
(b) **Special Linear Group**: $\mathrm{SL}(n, \mathbb{R}) = \{A \mid \det(A) = 1\} \leq \mathrm{GL}(n, \mathbb{R})$.
(c) **Roots of Unity**: $U_n = \{z \in \mathbb{C} \mid z^n = 1\} \leq U = \{z \in \mathbb{C} : |z| = 1\} \leq \mathbb{C}^\times$.

Definition 0.1.12. Cyclic Subgroup and Generator

For $a \in G$, the **cyclic subgroup generated by** a is

$$\langle a \rangle = \{a^n \mid n \in \mathbb{Z}\} \leq G \quad \text{(in additive notation, } \{na \mid n \in \mathbb{Z}\}\text{)}.$$

G is **cyclic** if $\exists a \in G$ such that $G = \langle a \rangle$; a is called a **generator** of G.

Theorem 0.1.13. $G = \langle a \rangle \implies G$ is abelian.

Theorem 0.1.14 (Division Algorithm for $\mathbb{Z}$)**.**

$$\forall m \in \mathbb{Z}^+, \forall n \in \mathbb{Z}, \quad \exists!\, q, r \in \mathbb{Z} \text{ such that } n = mq + r \quad \text{and} \quad 0 \leq r < m.$$

Theorem 0.1.15 (Subgroups of Cyclic Groups)**.** Let $G = \langle a \rangle$.

(a) $H \leq G \implies H$ is cyclic.
(b) The subgroups of $(\mathbb{Z}, +)$ are precisely $n\mathbb{Z} = \langle n \rangle$ for $n \in \mathbb{Z}_{\geq 0}$.

Definition 0.1.16. Greatest Common Divisor

For $r \in \mathbb{Z}^+, s \in \mathbb{Z}_{\geq 0}$, $\gcd(r, s)$ is the unique positive generator of $r\mathbb{Z} + s\mathbb{Z}$:

$$\langle \gcd(r,s) \rangle = \{nr + ms \mid n, m \in \mathbb{Z}\} \implies \exists n, m \in \mathbb{Z} \colon \gcd(r,s) = nr + ms.$$

Theorem 0.1.17 (Classification of Cyclic Groups)**.** Let $G = \langle a \rangle$.

(a) $|G| = \infty \implies G \cong (\mathbb{Z}, +)$ via $a^k \mapsto k$.
(b) $|G| = n < \infty \implies G \cong (\mathbb{Z}_n, +_n)$ via $a^k \mapsto k \pmod{n}$.

Theorem 0.1.18 (Orders, Generators, and Subgroup Lattice)**.** Let $G = \langle a \rangle$ with $|G| = n$, and let $b = a^s \in G$.

(a) **Element Order**:

$$|a^s| = |\langle a^s \rangle| = \frac{n}{\gcd(n, s)}, \quad \text{and} \quad \langle a^s \rangle = \langle a^{\gcd(n,s)} \rangle.$$

(b) **Generators**:

$$\langle a^s \rangle = G \iff \gcd(n, s) = 1.$$

The number of distinct generators of G is $\phi(n) = |\{k \in \{1, \ldots, n\} \mid \gcd(n, k) = 1\}|$.

(c) **Lagrange Divisibility**: $H \leq G \implies |H| \mid n$.

(d) **Subgroup Uniqueness**: For each divisor $d \mid n$ $(d > 0)$, there exists a **unique** subgroup of order d:

$$H_d = \langle a^{n/d} \rangle = \left\{x \in G \mid x^d = e\right\}, \quad |H_d| = d.$$

Theorem 0.1.19 (Subgroup Intersections)**.** Let $\{H_i\}_{i\in I}$ be any nonempty collection of subgroups of G. Then

$$\bigcap_{i\in I} H_i \leq G.$$

Definition 0.1.20. Subgroup Generated by a Subset

For $\varnothing \neq S \subseteq G$:

$$\langle S\rangle = \bigcap \{H \leq G \mid S \subseteq H\}.$$

S is a **generating set** of G if $\langle S\rangle = G$. If $\exists S \subseteq G$ with $|S| < \infty$ and $\langle S\rangle = G$, G is **finitely generated**.

Theorem 0.1.21 (Constructive Characterization)**.** For $\varnothing \neq S \subseteq G$:

$$\langle S\rangle = \{x_1^{\epsilon_1} x_2^{\epsilon_2} \cdots x_k^{\epsilon_k} \mid k \in \mathbb{N},\, x_j \in S,\, \epsilon_j \in \{\pm 1\}\}.$$

Definition 0.1.22. Cayley Digraph

Let G be a finite group and $S \subseteq G$ a generating set. The **Cayley digraph** $\Gamma(G,S) = (V,E)$ is defined by:

- $V = G$;
- $E = \{(g, gs; s) \mid g \in G,\, s \in S\}$, representing a directed arc $g \xrightarrow{s} gs$.

When $s = s^{-1}$, (g, gs) and (gs, g) collapse to an undirected edge.

Proposition 0.1.23 (Structural Properties of $\Gamma(G,S)$)**.**

(a) **Connectivity**: $\Gamma(G,S)$ is connected $\iff \langle S\rangle = G$.
(b) **Valency**: $\forall g \in G$, $\forall s \in S$, $\deg_{\text{in}}^{(s)}(g) = \deg_{\text{out}}^{(s)}(g) = 1$.
(c) **Vertex Transitivity**: $G \leq \text{Aut}(\Gamma(G,S))$ via left regular action $\lambda_h(g) = hg$.
(d) **Cycles and Relations**: Directed closed paths from e:

$$e \xrightarrow{s_1^{\epsilon_1}} \cdots \xrightarrow{s_k^{\epsilon_k}} e \iff s_1^{\epsilon_1} \cdots s_k^{\epsilon_k} = e \text{ in } G.$$

Example 0.1.24. Below is a Cayley digraph of D_4 with generating set $S = \{\rho, \mu\}$

Exercise 0.1.25. Let G be an arbitrary group and let $a, b \in G$.

- Prove that if $|ab| = n < \infty$, then $|ba| = n$.

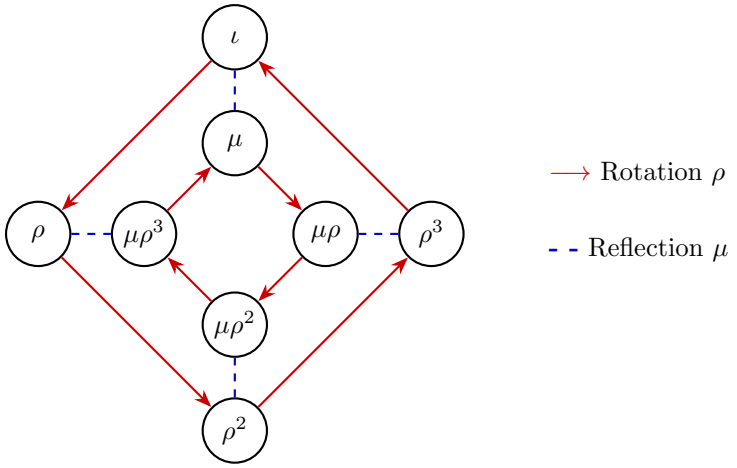
  Proof. Notice

  $$(ab)^n = a(ba)^{n-1}b = e \implies a(ba)^n = ea = a \implies (ba)^n = e \implies |ba| \leq n$$

  and

  $$((ba)^m = e \implies (ab)^m = e) \implies |ba| \geq |ab| = n$$

  Then $|ba| = n$. □



- Deduce that for any $g \in G$ and $x \in G$, $|x| = |gxg^{-1}|$

  Proof. We have

  $$(gxg^{-1})^m = gx^m g^{-1} = e \implies gx^m = g \implies x^m = e \implies \left|gxg^{-1}\right| \geq |x|$$

  We also have

  $$x^m = e \implies gx^m g^{-1} = e \implies gxg^{-1})^m = e \implies |x| \geq \left|gxg^{-1}\right|$$

  Therefore, $|x| = |gxg^{-1}|$. □

  Remark. In fact, let $a = gx$ and $b = g^{-1}$ then $|ab| = |gxg^{-1}| = |ba| = |x|$.

Exercise 0.1.26. Let H and K be subgroups of a group G, and define the subset $HK = \{hk : h \in H, k \in K\}$

- Prove that if G is abelian, then $HK \leq G$.

  Proof. Let $hk, h'k' \in HK$, then we have

  $$(hk)(h'k')^{-1} = hkk'^{-1}h'^{-1} = (hh'^{-1})(kk'^{-1}) \in HK$$

  implying that $HK \leq G$. □

- Give a counterexample when G is nonabelian.

  Proof. Let $G = D_3, H = \langle\mu\rangle, K = \langle\mu\rho\rangle$, then $HK = \{e, \mu, \rho, \mu\rho\}$ is not a subgroup of G. □

Exercise 0.1.27. Let G be a finite cyclic group. Prove that if $H, K \leq G$ with $H \neq K$, then $|H| \neq |K|$.

Proof. Let $G = \langle a\rangle$ and $|G| = n$. Then $H, K \leq G$ implies $H = \langle a^p\rangle = \langle a^{\gcd(n,p)}\rangle, K = \langle a^q\rangle = \langle a^{\gcd(n,q)}\rangle$ for some $p, q \in \mathbb{Z}^+$, then

$$H \neq K \implies \gcd(n,p) \neq \gcd(n,q) \implies |H| = \frac{n}{\gcd(n,p)} \neq \frac{n}{\gcd(n,q)} = |K|$$

□

Exercise 0.1.28. Prove that a group G with $|G| > 1$ has no proper nontrivial subgroups if and only if G is finite and is isomorphic to $\mathbb{Z}_p$ for some prime p.

Proof. $\implies$: Let $a \neq e \in G$.. Then $\langle a \rangle \leq G$ must equals G for otherwise it will be a proper nontrivial subgroup of G. Then G is cyclic. If G is infinite, then $\langle a \rangle = \langle a^2 \rangle = G$ leads to a contradiction since $a \notin \langle a^2 \rangle$. If $|G| = mn$ for some $m, n > 1$, then $|a^m| = mn/\gcd(mn, m) = n$ implying that $\langle a^m \rangle$ is a proper nontrivial subgroup of G. Therefore, $G = \langle a \rangle$ and $|G| = p$ with p prime. It follows that $G \cong \mathbb{Z}_p$.

$\impliedby$: Let $G \cong \mathbb{Z}_p$, then $G = \langle a \rangle$ and $|G| = p$. If $H \leq G$, then $H = \langle a^s \rangle = \langle a^{\gcd(p,s)} \rangle$. Since p is prime, we know $H = \{e\}$ or $H = G$. □

0.2 Finitely Generated Abelian Groups

Definition 0.2.1. Direct Product of Groups

Let $G_1, G_2, \ldots, G_n$ be groups. The **direct product** $\prod_{i=1}^n G_i = G_1 \times G_2 \times \cdots \times G_n$ is the Cartesian product endowed with componentwise binary operation:

$$(a_1, a_2, \ldots, a_n)(b_1, b_2, \ldots, b_n) = (a_1 b_1, a_2 b_2, \ldots, a_n b_n).$$

The identity element is $(e_1, e_2, \ldots, e_n)$, and $(a_1, \ldots, a_n)^{-1} = (a_1^{-1}, \ldots, a_n^{-1})$. It therefore forms a group.

Theorem 0.2.2. The group $\prod_{i=1}^n \mathbb{Z}_{m_i}$ is cyclic and isomorphic to $\mathbb{Z}_{m_1 m_2 \cdots m_n}$ if and only if $\gcd(m_i, m_j) = 1$ for all $i \neq j$.

Theorem 0.2.3. Let $(a_1, a_2, \ldots, a_n) \in \prod_{i=1}^n G_i$. If each a_i has finite order r_i in G_i, then

$$|(a_1, a_2, \ldots, a_n)| = \operatorname{lcm}(r_1, r_2, \ldots, r_n).$$

Theorem 0.2.4 (**Fundamental Theorem of Finitely Generated Abelian Groups - Primary Factor Form**)**.** Every finitely generated abelian group G is isomorphic to a direct product of cyclic groups:

$$G \cong \mathbb{Z}_{(p_1)^{r_1}} \times \mathbb{Z}_{(p_2)^{r_2}} \times \cdots \times \mathbb{Z}_{(p_n)^{r_n}} \times \underbrace{\mathbb{Z} \times \cdots \times \mathbb{Z}}_{\beta \text{ copies}},$$

where p_i are primes (not necessarily distinct) and $r_i \in \mathbb{Z}^+$. β is called the **Betti number** of G.

Theorem 0.2.5 (**Fundamental Theorem of Finitely Generated Abelian Groups - Invariant Factor Form**)**.** Every finitely generated abelian group G is isomorphic to a direct product of cyclic groups:

$$G \cong \mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2} \times \cdots \times \mathbb{Z}_{d_k} \times \underbrace{\mathbb{Z} \times \cdots \times \mathbb{Z}}_{\beta \text{ copies}},$$

where each $d_i \geq 2$ and $d_1 \mid d_2 \mid \cdots \mid d_k$. The **invariant factors** $d_1, \ldots, d_k$ and the Betti number β are uniquely determined.

Definition 0.2.6. Decomposable and Indecomposable Groups

A group G is **decomposable** if it is isomorphic to the direct product of two proper nontrivial subgroups: $G \cong H \times K$ with $H, K \neq \{e\}$. Otherwise, G is **indecomposable**.

Theorem 0.2.7.

$$G \text{ is finite indecomposable abelian } \iff G \cong \mathbb{Z}_{p^r} \text{ for some prime } p \text{ and } r \in \mathbb{Z}^+.$$

Theorem 0.2.8. If G is a finite abelian group and $m \mid |G|$, then G contains a subgroup of order m.

Theorem 0.2.9. If $m \in \mathbb{Z}^+$ is square-free, then every abelian group of order m is cyclic.

Exercise 0.2.10. Let $G = \mathbb{Z}_4 \times \mathbb{Z}_{18} \times \mathbb{Z}_{15}$ and $H = \mathbb{Z}_3 \times \mathbb{Z}_{36} \times \mathbb{Z}_{10}$. Are they isomorphic?

Proof. We have

$$G \cong \mathbb{Z}_4 \times (\mathbb{Z}_2 \times \mathbb{Z}_9) \times (\mathbb{Z}_3 \times \mathbb{Z}_5)$$

and

$$H \cong \mathbb{Z}_3 \times (\mathbb{Z}_4 \times \mathbb{Z}_9) \times (\mathbb{Z}_2 \times \mathbb{Z}_5)$$

It follows that $G \cong H$. □

Exercise 0.2.11. Let G be a finite abelian group. Prove that G is not cyclic if and only if G contains a subgroup isomorphic to $\mathbb{Z}_p \times \mathbb{Z}_p$ for some prime p.

Proof. Since G is finite abelian, then

$$G \cong \prod_{i=1}^{n} \mathbb{Z}_{(p_i)^{r_i}}$$

By Theorem 0.2.2, if it is not cyclic, then $\gcd((p_i)^{r_i}, (p_j)^{r_j}) > 1$ for some $i \neq j$, implying that $p_i = p_j$ and then

$$\mathbb{Z}_{p_i} \times \mathbb{Z}_{p_i} \cong \langle (p_i)^{r_i - 1} \rangle \times \langle (p_j)^{r_j - 1} \rangle \leq \mathbb{Z}_{(p_i)^{r_i}} \times \mathbb{Z}_{(p_j)^{r_j}} \leq \prod_{i=1}^{n} \mathbb{Z}_{(p_i)^{r_i}} \cong G$$

If G is cylic, then $p_i \neq p_j$ for all $i \neq j$. It follows that if $H \leq G$, then

$$H \cong \prod_{i=1}^{n} Z_{(p_i)^{q_i}}, q_i \leq r_i$$

which is not isomorphic to $\mathbb{Z}_p \times \mathbb{Z}_p$ since p_i are mutually distinct. □

Remark. In fact, when G is cylic, we can deduce that $H \leq G$ is also cyclic, which directly tells proves the result.

Remark. Generally, one can prove that:

> Let G be a finite abelian group, and let $x_1, \ldots, x_n \geq 2$ be integers. The group $H = \prod_{i=1}^{n} \mathbb{Z}_{x_i}$ is isomorphic to a subgroup of G if and only if G can be decomposed as a direct product of cyclic groups $G \cong \prod_{j=1}^{m} \mathbb{Z}_{y_j}$ such that $m \geq n$ and $x_i \mid y_i$ for all $i \in \{1, \ldots, n\}$.

Exercise 0.2.12. Let G, H, K be finitely generated abelian groups. Show that

$$G \times K \cong H \times K \implies G \cong H$$

Proof. Since G, H, K are finitely generated abelian groups, so are $G \times K$ and $H \times K$. Since $G \times K \cong H \times K$, so they have the same Betti number and the same collection of $(p_i)^{r_i}$ after decomposition. It follows that after picking up the part given by K, we know G and H have the same Betti number and the same collection of $(p_i)^{r_i}$ after decomposition, which implies $G \cong H$. □

0.3 Cosets and the Theorem of Lagrange

Theorem 0.3.1. Let $H \leq G$. The relation $\sim_L$ defined on G by

$$a \sim_L b \iff a^{-1}b \in H$$

is an equivalence relation on G. Similarly, $a \sim_R b \iff ab^{-1} \in H$ is an equivalence relation.

Definition 0.3.2. Coset

Let $H \leq G$ and $a \in G$.

- The **left coset** of H containing a is the equivalence class under $\sim_L$:

$$aH = \{ah \mid h \in H\}.$$

- The **right coset** of H containing a is the equivalence class under $\sim_R$:

$$Ha = \{ha \mid h \in H\}.$$

Theorem 0.3.3. Let $H \leq G$. For every $a \in G$, the map $f \colon H \to aH$ defined by $f(h) = ah$ is a bijection. Consequently,

$$|aH| = |H| = |Ha| \quad (\forall a \in G).$$

Theorem 0.3.4 (Theorem of Lagrange)**.** Let H be a subgroup of a finite group G. Then the order of H divides the order of G:

$$|H| \mid |G|.$$

Corollary 0.3.5. Every group of prime order p is cyclic, hence isomorphic to $\mathbb{Z}_p$.

Theorem 0.3.6. Let G be a finite group. For every $g \in G$, $|g|$ divides $|G|$. In particular, $g^{|G|} = e$.

Definition 0.3.7. Index of a Subgroup

Let $H \leq G$. The **index** of H in G, denoted $(G : H)$ or $[G : H]$, is the cardinality of the set of left (or right) cosets of H in G. When G is finite,

$$[G : H] = \frac{|G|}{|H|}.$$

Theorem 0.3.8 (Tower Law for Subgroup Indices)**.** Let $K \leq H \leq G$. If $[G : H]$ and $[H : K]$ are finite, then $[G : K]$ is finite and

$$[G : K] = [G : H] \cdot [H : K].$$

Theorem 0.3.9 (Cosets of Homomorphism Kernels)**.** Let $\phi\colon G \to G'$ be a group homomorphism. Then

(a) Left and right cosets of $\ker(\phi)$ coincide:

$$a\ker(\phi) = \ker(\phi)a \quad (\forall a \in G).$$

(b) Two elements $a, b \in G$ belong to the same coset of $\ker(\phi)$ if and only if $\phi(a) = \phi(b)$:

$$a\ker(\phi) = b\ker(\phi) \iff \phi(a) = \phi(b).$$

Corollary 0.3.10. A group homomorphism $\phi\colon G \to G'$ is injective if and only if

$$\ker(\phi) = \{e_G\}.$$

Exercise 0.3.11. Let $\sigma = (1, 2, 5, 4)(2, 3)$ in S_5, find the index of $\langle\sigma\rangle$ in S_5.

Proof. Notice $\sigma = (1, 2, 3, 5, 4)$, then the order of σ is 5 in S_5, implying the index given by

$$[S_5 : \langle\sigma\rangle] = \frac{|S_5|}{|\langle\sigma\rangle|} = \frac{5!}{5} = 24$$

□

Exercise 0.3.12. Let H be a subgroup of a group G, let $a, b \in G$. Is it true that

$$aH = bH \implies Ha^{-1} = Hb^{-1}$$

Proof. The statement is true. Since $aH = bH$, then $ah_a = bh_b$ for some $h_a, h_b \in H$, it follows that $b^{-1}a \in H$. Therefore, for $hb^{-1} \in Hb^{-1}$, we have $hb^{-1} = hb^{-1}aa^{-1} = (hb^{-1}a)a^{-1} \in Ha^{-1}$. Then $Hb^{-1} \subset Ha^{-1}$, similarly we can deduce that $H^{-1} \subset Hb^{-1}$, thus $Ha^{-1} = Hb^{-1}$. □

Exercise 0.3.13. Let G be a group of order pq, where p and q are prime numbers. Show that every proper subgroup of G is cyclic.

Proof. By Theorem of Lagrange we know every proper subgroup H has order $|H| = n \mid pq$ implying $n = 1$ or $n = p$ or $n = q$. If $n = 1$ then H is trivially cyclic, if $n = p$ or $n = q$ then Corollary 0.3.5 tells that H is cyclic. □

1 Glossary

Abstract Algebra Prerequisites

Group

Homomorphism and Isomorphism

Permutation and Symmetric Group

Cycle and Transposition

General Linear Group

Dihedral Group

Subgroup

Cyclic Subgroup and Generator

Greatest Common Divisor

Subgroup Generated by a Subset

Cayley Digraph

Direct Product of Groups

Decomposable and Indecomposable Groups

Coset

Index of a Subgroup